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# URA302 – WORK SHEET 5
# NumPy, SciPy, Matplotlib
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import numpy as np
import matplotlib.pyplot as plt
from scipy import stats, fftpack, interpolate, signal
from scipy.linalg import det, inv, eig
from scipy.optimize import curve_fit, minimize
from scipy.integrate import odeint
import time, random, string, os

# -------------------------
# Q1: Statistical Properties
# -------------------------
print("\nQ1:")
arr = np.random.rand(10)
print("Array:", arr)
print("Mean:", np.mean(arr))
print("Median:", np.median(arr))
print("Variance:", np.var(arr))

# -------------------------
# Q2: 2D FFT
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print("\nQ2:")
A = np.random.rand(4,4)
fft2d = fftpack.fft2(A)
print("2D FFT:\n", fft2d)

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# Q3: Linear Algebra
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print("\nQ3:")
M = np.array([[4, 2], [3, 1]])
print("Determinant:", det(M))
print("Inverse:\n", inv(M))
print("Eigenvalues:", eig(M)[0])

# -------------------------
# Q4: Interpolation
# -------------------------
print("\nQ4:")
x = np.linspace(0,10,10)
y = np.sin(x)
f = interpolate.interp1d(x, y, kind='cubic')
xnew = np.linspace(0,10,100)
plt.figure()
plt.plot(x, y, 'o', label="Data")
plt.plot(xnew, f(xnew), label="Interpolated")
plt.legend()
plt.title("Interpolation")
plt.show()

# -------------------------
# Q5: Signal Filtering
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print("\nQ5:")
t = np.linspace(0, 1, 500)
sig = np.sin(2*np.pi*50*t) + np.random.normal(0,0.5,500)
filtered = signal.savgol_filter(sig, 51, 3)
plt.figure()
plt.plot(t, sig, label="Noisy")
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plt.plot(t, filtered, label="Filtered")
plt.legend()
plt.show()

#
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